

Nonlinear Acoustic Feedback in Architectural Spaces: Harnessing Resonant Anomalies for Generative Sound Systems

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ABSTRACT

Acoustic feedback is traditionally treated in audio engineering as an unwanted failure state to be prevented by narrow-band notch filters and automatic gain attenuation. In this treatise, we reframe controlled positive acoustic feedback as a vital, non-linear musical instrument. By coupling directional microphone arrays to architectural room modes and governing amplifier gain with dynamic hysteresis limiters, sound systems can generate self-regulating polyphonic resonance without manual intervention. We present mathematical models of acoustic room gain, physical placement criteria for resonant balance, and hardware schematics for real-time phase compensation.

KEY THEOREMS & BOUNDARY EQUATIONS

Formulation [1.1]

Theorem of Room Mode Equilibrium: In any enclosed space with $RT60 > 1.2s$, closed-loop gain can be dynamically stabilized by phase-inverting secondary standing nodes without muting primary harmonics.

Formulation [2.1]

The Hysteresis Damping Principle: Introducing non-linear saturation into the amplifier feedback loop prevents runaway squeal while sustaining warm, continuously evolving harmonic overtones.

Formulation [3.1]

Spatial Feedback Polyphony: By distributing multiple microphone-speaker pairs across an asymmetric room, independent acoustic feedback voices can coexist without phase cancellation.

1. Introduction: From Artifact to Instrument

Every enclosed architectural space possesses a unique resonant fingerprint—a set of standing frequencies governed by its physical dimensions, wall materials, and room geometries. In standard sound reinforcement, acoustic feedback occurs when energy from a speaker returns to a microphone with unity or greater gain, rapidly escalating into destructive distortion. However, when viewed through the lens of cybernetic systems and experimental composition, feedback is a pure dialogue between physical architecture and electroacoustic transduction.

2. The Hysteresis Gain Architecture

To transform runaway screech into melodic resonance, we developed an analog dynamic limiter utilizing soft-clipping JFET stages and optical photoresistors. As acoustic energy approaches room saturation, the circuit softly compresses the primary peak while exciting odd harmonics. The result is a lush, self-modulating acoustic organ that breathes with atmospheric air currents, human movement, and room temperature shifts.

3. Spatial Staging and Multi-Node Calibration

Deploying feedback systems across large public spaces requires careful spatial staging. By arranging directional hypercardioid microphones in opposing quadrants and feeding adjacent speaker clusters through fractional delays (12–48 ms), we decouple room nodes into distinct vibrational zones. Performers can interact with the sound field simply by moving their bodies between reflection points, dampening or releasing specific harmonics.

4. Conclusion and Artistic Applications

The ZIAA feedback installations demonstrate that acoustic anomalies are not engineering flaws to be eliminated, but rich expressive resources. Future work focuses on linking real-time room feedback to algorithmic microtonal synthesizers, allowing architectural spaces to autonomously compose their own generative soundscapes.

REFERENCES & PRIMARY CITATIONS

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